

Information Collection

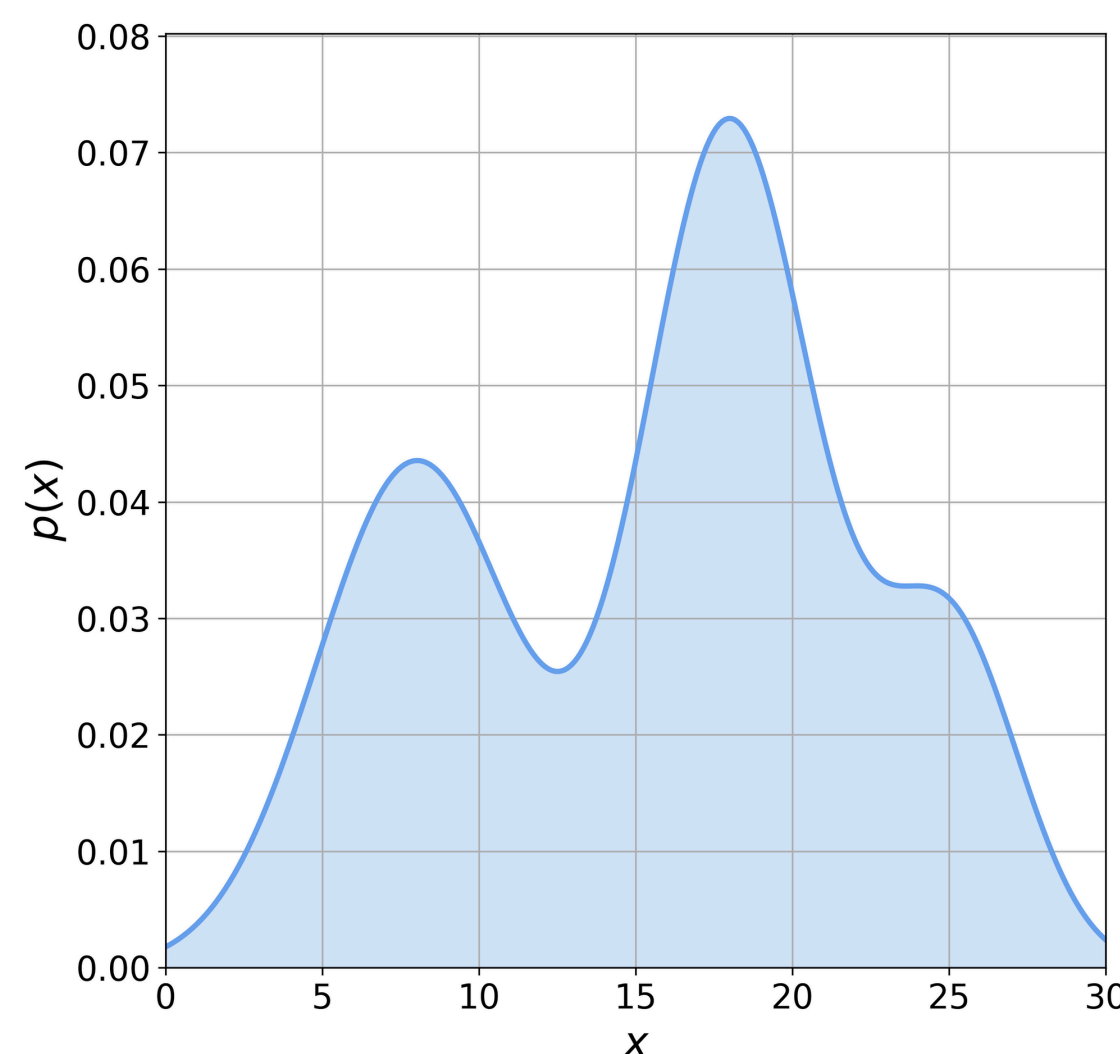
DATA and COMPOSITE

Find, identify and meaningfully propagate uncertainties from data sources

QUALITATIVE

Multiple opinions to obtain a discrete set of valid options

If data is uncertain, it is presented as a distribution



$$\mathbb{D} = \{f_{c_1}(x), \dots, f_{c_n}(x)\}$$

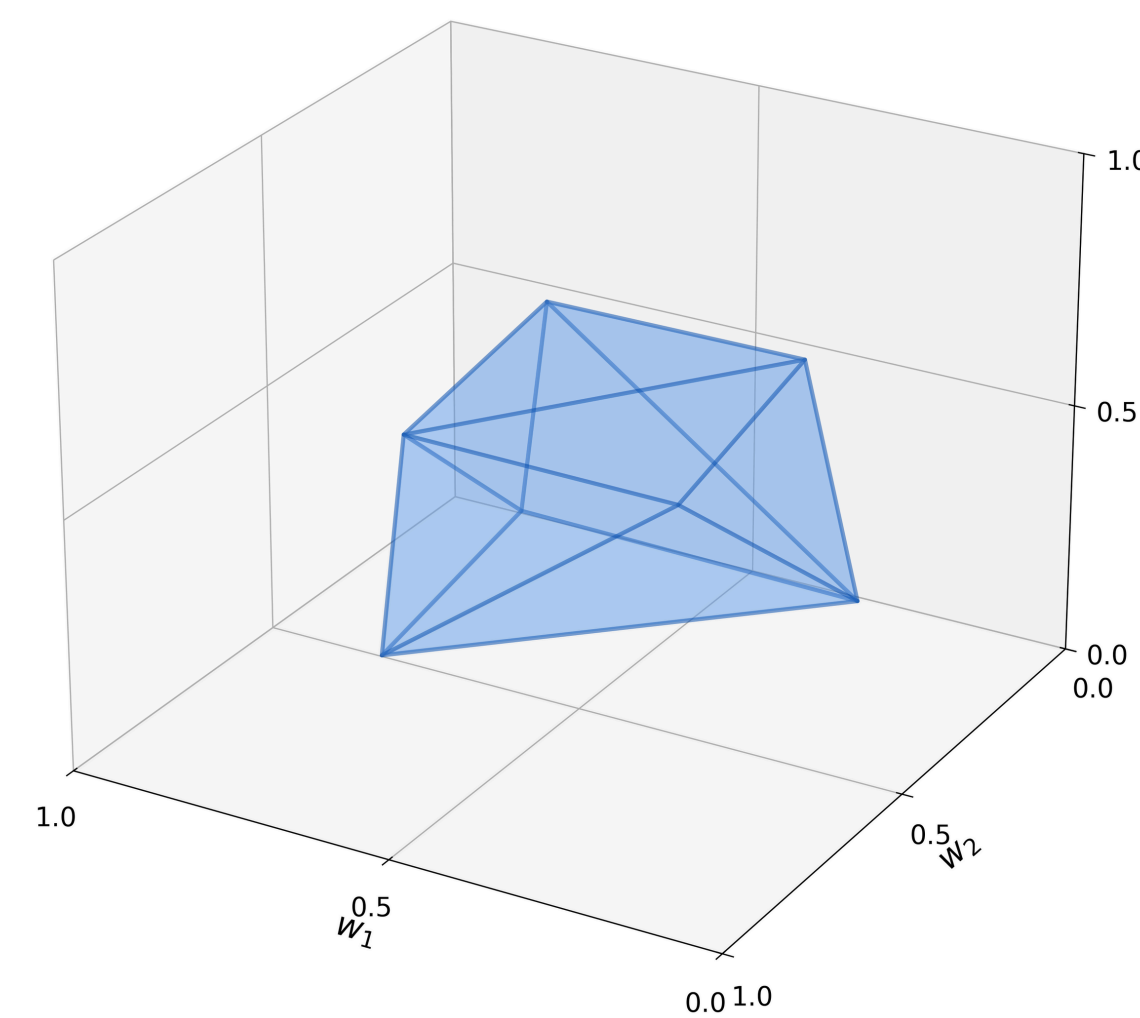
Set of input data distributions, one per each indicator

Elicitation sessions

PILE-BWT

Pairwise comparisons, solved by optimization

All the weight sets from the weight space are valid

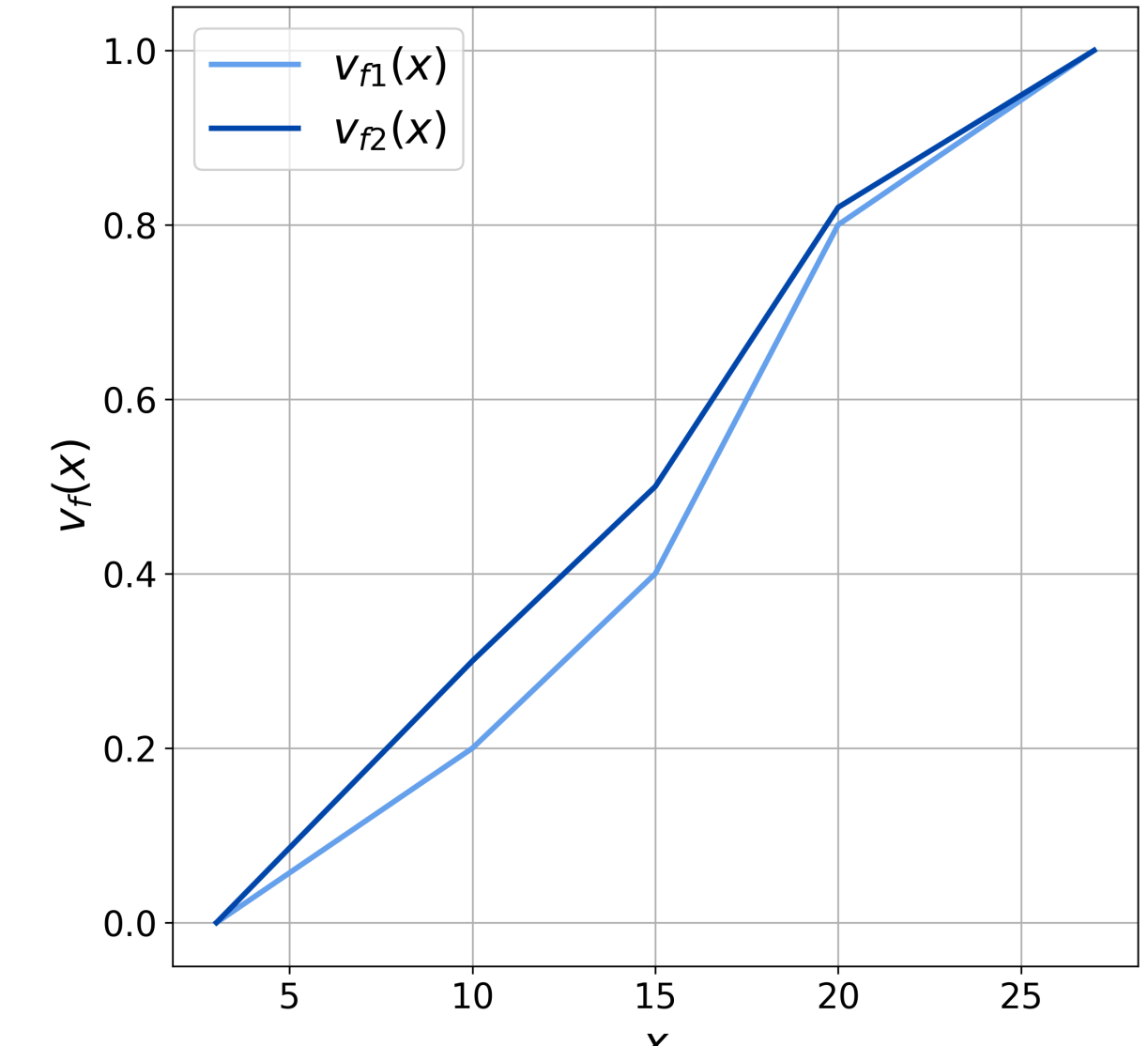


$$\mathbb{W} = \{\underline{w}_a, \underline{w}_b\}$$

Set of allowable weight spaces, one per each opinion

Mid-point splitting method and direct elicited corrections

Every value function is valid and might be considered



$$\mathbb{V} = \{\underline{v}_a, \underline{v}_b\}$$

Set of allowable value functions one per each opinion

Strict MC

For each opinion (a)

Montecarlo Run

For each Alternative (α)

Sample a set of weight "w" from the allowable weight space

$$\underline{w}_\alpha \in \underline{w}_a \in \mathbb{W}$$

For each criterion (i)

Sample data point "d ∈ D"

$$\mathbb{D} \ni f_i(x) \rightarrow d_i$$

Select value function "v = V_e,c"

$$v_i(x) \in \underline{v}_a \in \mathbb{V}$$

Aggregate criteria values to get alternative values

$$A_\alpha = \sum_i^c w_i \cdot v_i(d_i)$$

$$\underline{A}_a = \{A_\alpha, \dots\}$$

Set of alternative value distributions for opinion a, one for each MC run

Non-Strict MC

Montecarlo Run

For each Alternative (α)

Sample an opinion a ∈ {a, ...}

Sample a set of weight "w" from the allowable weight space

$$\underline{w}_\alpha \in \underline{w}_a \in \mathbb{W}$$

For each criterion (i)

Sample data point "d ∈ D"

$$\mathbb{D} \ni f_i(x) \rightarrow d_i$$

Sample an opinion a* ∈ {a, ...}

Select value function "v = V_e,c"

$$v_i(x) \in \underline{v}_{a^*} \in \mathbb{V}$$

Aggregate criteria values to get alternative values

$$A_\alpha = \sum_i^c w_i \cdot v_i(d_i)$$

$$\underline{A} = \{A_\alpha, \dots\}$$

Set of alternative value distributions, one for each MC run

Distributions of value for each alternative for each opinion

$$\{\underline{A}_a, \dots\}$$

UNCERTAINTY ANALYSIS

CONSENSUS ANALYSIS

Distributions of value for each alternative, aggregated opinions

$$\underline{A}$$

RANKING HEATMAP